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Experimental and kinetic modeling investigation on laminar flame propagation of CH_4/CO mixtures at various pressures: Insight into the transition from CH_4-related chemistry to CO-related chemistry

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ABSTRACT

In this work, laminar flame speeds of CH_4/CO/air mixtures were measured at the unburnt temperature of 353 K and pressures from 1 to 10 atm in a high-pressure constant-volume cylindrical combustion vessel. Effects of pressure, equivalence ratio and CO content in CH_4/CO mixtures on laminar flame speeds were investigated. A kinetic model for CH_4/CO combustion was developed based on recent progress in elementary reactions and validated against previous and present experimental targets. It is found that both the thermal effect originating from different adiabatic flame temperatures and chemical effect originating from differences in the radical pool play important roles in the variation of laminar flame speed. The separate contribution of each effect varies at different pressures, equivalence ratios and CO contents. Besides, the transition from CH_4-related chemistry to CO-related chemistry can be monitored by the increasing concentration of O atom in the radical pool under all the investigated conditions. Based on the modeling analysis, R18 ($CH_3 + CH_3$ (+M) = C_2H_6 (+M)), R19 (HCO (+M) = H + CO (+M)), R6 (CH4 (+M) = CH_3 + H (+M)) and R7 (CH_4 + H = $CH_3 + H_2$) have major contributions to the transition chemistry, especially under the rich conditions. Compared with the atmospheric pressure conditions, R20 (CO + O (+M) = CO_2 (+M)) and R22 (H + O_2 (+M) = HO_2 (+M)) are enhanced at high pressures, which leads to the decrease of O atom in the radical pool at high pressures.

1. Introduction

The combustion of fossil fuels has led to rapid fossil energy consumption and serious environmental challenges. To this end, two roadmaps are currently considered to meet these challenges: developing clean and renewable energy sources or improving the combustion efficiency. Biomass ranks fourth as an energy source worldwide which can be converted into liquid, solid and gasified fuels [1]. Among them, biomass derived gas (BDG), which mainly contains CO, H_2, CH_4, N_2 and CO_2 [2,3], is attractive and versatile because of its wide availability and potential to reduce CO_2 emission. However, the lower heating value of BDG is relatively low, which may require co-firing of BDG with other fuels such as natural gas (NG) to increase the energy density of the gas. Besides, in the application of internal combustion engines, the addition of reformed gas like H_2, syngas or CO to NG could increase the flame speed [4] and improve the engine thermal efficiency [5]. Therefore, understanding the combustion chemistry of BDG and NG mixtures is essential to develop the combustion technologies and improve combustion efficiency. In order to isolate the effects of mixture composition, in this case, two of the major components, CO and CH_4 are selected as target fuels.

As one of the most important global combustion parameters, laminar flame speed is not only important for the determination of turbulent flame speed but also very helpful for the validation of the high-temperature reaction mechanism. Scholte and Vaags measured the laminar flame speeds of CH_4/CO/air mixtures at ambient temperature and pressure [6]. According to their experimental results, the laminar flame speed first increases slowly with the increase of CO and then sharply decreases. The peak value locates at 90% of CO content in the CO/CH_4 mixture. With the development of experimental methods, Vagelopoulos and Egolfopoulos [7] also investigated the laminar flame propagation of CH_4/CO mixtures at 298 K and 1 atm and their measured results are notably lower than those obtained by Scholte and Vaags [6]. However, the same trend was observed in their experiments. Based on modeling analysis, they concluded that the addition of small amount of CH_4 to

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CO flame could accelerate the main CO oxidation reaction, while for large amount of CH_4 addition, the chemistry shifts toward that of CH_4. Subsequent laminar flame speed measurements of CO/CH_4 mixtures were conducted by Konnov et al. [8] at 298 K and 1 atm with the CO content varying from 0% to 15%. In their experiments, only a slight increase of the laminar flame speeds was observed when CO was added, which was in good agreement with previous measurements [6,7]. Recently, He et al. [9] investigated the effects of CO addition on $CH_4/CO_2/N_2$ mixture at 298 K and 1 atm. The measured peak laminar flame speed locates at 80% of CO content in the CO/CH_4 mixture. Based on modeling analysis, they concluded that this phenomenon was caused by the change in consumption rates of CO and OH and the discrepancy of the heat release rate in the preheat zone.

Besides these experimental studies, Sung et al. [10] investigated the effects of CO addition on the laminar flame speed of *n*-butane/air mixtures numerically. The CO contents in the mixtures were 10% and 20%. They found the increase of the laminar flame speed with CO addition is mainly due to thermal effect, especially under the lean conditions. Liu et al. [11] investigated the effect of CO addition in syngas/CH_4 mixtures with the CO contents equal to 15% and 30% numerically. They found around 75% of the increase in laminar flame speed is due to thermal effect under lean and stoichiometric conditions. In contrast, numerical simulation work performed by Wu et al. [12] concluded that the effect of CO addition on the laminar flame speed of the stoichiometric CH_4/CO/air mixture was dominated by the chemical effect of the transition of dominant reaction steps. Besides, they also highlighted the contradictory to the conclusion of Sung et al. [10] that the effect of CO addition on the laminar flame speed is thermal in nature.

In summary, based on previous studies, the trend of laminar flame speeds versus CO content in the CH_4/CO mixture is in good consistency, which first increases and then decreases rapidly. This phenomenon was proposed to be driven by the transition chemistry from CH_4 to CO. However, these studies were all investigated at ambient temperature and pressure. The effect of CO addition is still in debate and the transition from CH_4-related chemistry to CO-related chemistry has not been clarified under various conditions. In this work, laminar flame speeds of CH_4/CO/air were measured at both the lean and rich conditions, extending previous atmospheric pressure conditions to 10 atm. In order to reduce the flame instabilities at high pressures, oxygen and helium were also considered as the oxidizer. Therefore, laminar flame speeds of $CH_4/CO/O_2$/He mixtures were also measured at 5 and 10 atm. The CO content in CH_4/CO mixtures was varied from 0% to 95% and the initial temperature was kept at 353 K. A kinetic model for the combustion of CH_4/CO mixture was developed based on recent progress in the elementary reactions and validated against previous and present experimental targets. Based on the present model, the transition from CH_4-related chemistry to CO-related chemistry is clearly characterized under different equivalence ratios and pressures. The thermal and chemical effects of CO addition under various conditions are also analyzed in response to the contradictory conclusions between Wu et al. [12] and Sung et al. [10].

2. Experimental method

The experiments were carried out in a high-pressure constant-volume cylindrical combustion vessel with the inner volume of 2.77 L. Details about the combustion vessel can be found elsewhere [13]. Premixed combustible mixtures of CH_4/CO/oxidizer (O_2: 21%, N_2:79% or O_2: 15%, He: 85%) were prepared in a premixing vessel in order to keep identical mixture composition for one specific equivalence ratio. Measurements were repeated 3–4 times for each experimental condition. The temperatures of the combustion vessel, the premixing vessel and the gas transferring pipelines were kept at 353 K in order to ensure the availability of experiments at elevated pressures. The combustible mixture was ignited by the discharging of two electrodes fixed at the center of the combustion vessel. The propagation of the spherical flame was recorded by a high speed camera (Phantom V310) via a schlieren system which was built up through two 75-mm-diameter apertures in the combustion vessel. The spatial resolution of the high-speed camera was set at 480×480 pixels and the recording speed was set at 12,001 frames/s.

Carbon monoxide is tend to react with iron in high-pressure steel cylinders to produce iron pentacarbonyl which can bring additional experimental uncertainties [14]. To avoid this, the carbonyl-free carbon monoxide source (purity: 99.99%) was mixed with methane (purity: 99.99%) to obtain six synthetic CH_4/CO mixtures with CO content (αCO) of 20%, 40%, 60%, 80%, 90% and 95%. All these mixtures were supplied in pressurized aluminum cylinders with brass fittings. The combustible mixture was pumped out of the pipelines and vessels as soon as the experiments were finished. Therefore, the residence time of CO in the stainless steel containers is no longer than two hours. Das et al. [15] conducted the laminar flame speed measurements for 95%CO/5%H_2 by using a cold trap to remove pentacarbonyl and compared the results obtained from the experiments carried out without using a cold trap. Negligible difference between the two sets of results is found. The measurements for CH_4/CO/air mixtures were conducted at 1, 2, 5 and 10 atm while those for $CH_4/CO/O_2$/He mixtures were performed at 5 and 10 atm. The data processing method in this work is referred to the nonlinear extrapolation methods developed by Kelley and Law [16] and Halter et al. [17] and has been introduced in detail in our previous work [13]. Detailed descriptions on the uncertainty analysis can be found in the *Supplementary materials*. The absolute uncertainties of the measured laminar flame speeds are between 0.64–3.21 cm/s depending on the equivalence ratios and initial pressures. All the experimental data, as well as the experimental uncertainties can be found in the *Supplementary materials*.

3. Kinetic modeling

The present model was developed based on our previous methanol model [18]. Particularly, the chemically termolecular reactions (R1–R4) were newly incorporated in the present model based on recent theoretical calculation work [19]. Other reactions and their kinetic parameters involved in the base model were kept the same. The sub-mechanism of CO was mainly taken from the Li et al.'s model [20]. For R5, theoretically calculated rate constant by Joshi and Wang [21] was adopted. Their calculated rate constant is 13% lower than that evaluated by Li et al. [20].

$$H + O_2 + H = H_2 + O_2 \tag{R1}$$

$$H + O_2 + H = OH + OH \tag{R2}$$

$$H + O_2 + O = OH + O_2 \tag{R3}$$

$$H + O_2 + OH = H_2O + O_2 \tag{R4}$$

$$CO + OH = CO_2 + H \tag{R5}$$

3.1. The sub-mechanism of CH_4

For the sub-mechanism of methane, one of the most important reaction is the unimolecular decomposition reaction of CH_4 (R6). The experimental measurements for R6 obtained in shock tube are mainly available above 1800 K [22–29]. At the same time, the reverse reaction of R6, i.e. the recombination of $CH_3 + H$, was measured over the temperature range of 300–600 K [30,31]. However, the existing experimental data at intermediate temperatures (1000–1800 K) are quite deficient [32,33]. Recently, Wang et al. [34] reported an improved measurement for $CH_4 + Ar = CH_3 + H + Ar$ behind shock waves. The quantitatively resolved CH_3 time histories allowed extension of their previous measurement range [24,27] to lower temperatures (1487–1866 K) at 1.7 atm. Their measured results are in good agreement with various theoretical and review studies [35,36]. A re-evaluation of rate constant based on their experimental data and previous data in the falloff region was also performed between 1000 and 2500 K, yielding updated expressions for both the low-pressure limit and the high-pressure limit rate constants which have improved agreement with all existing data. This evaluation result was adopted in the present model. In addition, the collisional efficiencies of various colliders were taken from the calculation results of Jasper et al. [37–39]. In these studies, the collision efficiency was investigated for a series of colliders, including He, Ne, Ar, Kr, H_2, N_2, O_2, CH_4 and H_2O with various master equation models.

$$CH_4\ (+M) = CH_3 + H\ (+M) \tag{R6}$$

For the rate constant of the H-abstraction reaction of CH_4 by H (R7), the experimental and theoretical results are in good agreement above 1000 K while they present large discrepancies below 500 K. Sutherland et al. [40] measured the rate constants of R7 and also its reverse reaction covering 938–1806 K using a shock tube. Besides, they also evaluated the rate constant of R7 based on their own data and literature data over 348–1950 K. Their evaluation value is 12% lower and 20% larger than that evaluated by Baulch et al. [36] at 600 K and 2000 K, respectively. In the present model, the evaluation results by Sutherland et al. [40] were adopted. The differences among the recommended rate constants [36,41,42] of the H-abstraction reaction of CH_4 by O (R8) are within 50% over 700–2500 K. In the present model, the evaluated rate constant by Baulch et al. in 1992 [43] was adopted, which is 30% higher and 40% lower than those evaluated by Cohen [41] and Baulch et al. in 2005 [36], respectively, in the temperature range of 700–2500 K. For the H-abstraction reaction of CH_4 by OH (R9), there are 103 records of rate constant sources listed in the NIST database [44]. Among them, the measurements by Srinivasan et al. [45] were conducted over the temperature range of 840–2025 K in a shock tube. Besides, based on their own data and literature data, an evaluation for the rate constant of R9 was performed over the temperature range of 195–2025 K, which was adopted in the present model. For the H-abstraction reaction of CH_4 by HO_2 (R10), the studies on its rate constant are very limited. The most recent theoretical calculation work [46] was performed by using explicitly correlated coupled-cluster theory with singles and doubles (CCSD-R12) in a large 19s14p8d6f4g3h basis (9s6p4d3f for H) to approach the basis-set limit at the coupled-cluster singles-doubles level. Their calculated rate constant is close to that recommended by Baulch et al. [36], and also in good agreement with the measurements of Baldwin et al. [47]. Therefore, the calculated value from Aguilera-Iparraguirre et al. [46] was adopted in the present model.

$$CH_4 + H = CH_3 + H_2 \tag{R7}$$

$$CH_4 + O = CH_3 + OH \tag{R8}$$

$$CH_4 + OH = CH_3 + H_2O \tag{R9}$$

$$CH_4 + HO_2 = CH_3 + H_2O_2 \tag{R10}$$

3.2. The sub-mechanism of CH_3

The sub-mechanism of CH_3 mainly involves the reactions of $CH_3 + O_2$ and $CH_3 + H/O/OH/HO_2$. These reactions normally incorporate different branching channels. For example, the combination of CH_3 and O_2 mainly includes three reaction channels, i.e. R11–R13. A review on these reaction channels can be found in the work of Srinivasan et al. in 2005 [48]. Two years later, they re-investigated R11 by using a new reflected shock tube and re-evaluated its rate constant [49], which was adopted in the present model. For R12, the measured value from Srinivasan et al. in 2005 [48] was adopted in the present model. In addition, the rate constant of R13 was adopted from the work of Fernandes et al. [50]. The review on the rate constant of $CH_3 + O$ can be found in a recent theoretical work from Xu et al. [51]. Up to now, there is a general consensus among different studies [51–57] on the global rate constant of $CH_3 + O$ at different temperatures, implying it is a weak temperature-dependent reaction. In the present model, the calculated value from Harding et al. [57] was adopted. R14 and R15 are two important branching reactions and the branching ratio of R15 is still in debate to date. The kinetic measurement with laser flash photolysis and mass spectrometer by Hack et al. [52] in 2005 gave the branching ratio of R14 to be 0.45 ± 0.05 at ambient temperature, which is close to that reported in literature [51,58]. In the present model, the branching ratio of R15 was taken from Hack et al. [52]. For the reactions of $CH_3 + HO_2$, Jasper et al. [59] conducted a theoretical investigation and found R16 and R17 are two important reaction channels. The calculated rate constant for R16 is in good agreement with the measured result by Hong et al. [60], which was adopted in the present model. As for R17, the calculated result by Jasper et al. [59] in the forward direction also shows good agreement with previous measurements [60,61], which was adopted in the present model. For the reverse reaction of R17, recent theoretical calculation result from Mai et al. [62] was adopted, which is in good agreement with previous measured result [61]. Key reactions discussed in this work are listed in Table 1.

$$CH_3 + O_2 = CH_2O + OH \tag{R11}$$

$$CH_3 + O_2 = CH_3O + O \tag{R12}$$

$$CH_3 + O_2\ (+M) = CH_3O_2\ (+M) \tag{R13}$$

$$CH_3 + O = CH_2O + H \tag{R14}$$

$$CH_3 + O = CO + H + H_2 \tag{R15}$$

$$CH_3 + HO_2 = CH_3O + OH \tag{R16}$$

$$CH_3 + HO_2 = CH_4 + O_2 \tag{R17}$$

3.3. Simulation methods

In this work, the laminar flame speed simulations were performed with the Premixed Laminar Flame-Speed module in the Chemkin PRO software [68]. The simulations were converged to a grid-independent solution by assigning both GRAD and CURV values of 0.1. Besides, both the multi-averaged transport and the thermal diffusion effect were considered.

In this work, three recent models, i.e. the high-pressure C_0–C_2 model developed by Ju and coworkers (referred as the HP model) [69], the Foundational Fuel Chemistry Model developed by Wang and coworkers (referred as the FFCM-1 model) [70] and the San Diego model [71] developed in 2016 were also used for comparison. Besides the present experimental data, other validation data

Table 1
The reactions discussed in this work and their rate constants in the form of k=AT^{n}exp($-E_a$/RT). The units are K, s^{-1}, cm^3, and cal/mol.

No.	Reactions	A	n	E_a	Reference
R1	$H + O_2 + H = H_2 + O_2$	8.80×10^{22}	−1.835	800	[19]
R2	$H + O_2 + H = OH + OH$	4.00×10^{22}	−1.835	800	[19]
R3	$H + O_2 + O = OH + O_2$	7.35×10^{22}	−1.835	800	[19]
R4	$H + O_2 + OH = H_2O + O_2$	2.56×10^{-7}	−1.835	800	[19]
R5	$CO + OH = CO_2 + H$	7.05×10^{4}	2.05	−276	[63]
	Duplicate				
	$CO + OH = CO_2 + H$	5.76×10^{12}	−0.66	332	
	Duplicate				
R6	CH_4 (+M) = CH_3 + H (+M)	2.10×10^{16}	0.00	104,913.6	[34,37–39]
	Low /	3.91×10^{17}	0.00	89,812.4/	
	Troe /	0.50 1350 1350 7834/			
	H_2/4.0/ H_2O/8.0/ CO/2.3/ CO_2/4.0/ Ar/1.0/ CH_4/5.0/ C_2H_6/4.0/ N_2/1.5/ He/3.0/ CH_3OH/6.0/ O_2/1.5/				
R7	$CH_4 + H = CH_3 + H_2$	4.08×10^{3}	3.156	8755.6	[40]
R8	$CH_4 + O = CH_3 + OH$	6.92×10^{8}	1.56	8484.49	[43]
R9	$CH_4 + OH = CH_3 + H_2O$	1.00×10^{6}	2.182	2446	[45]
R10	$CH_4 + HO_2 = CH_3 + H_2O_2$	1.13×10^{1}	3.74	21,010	[46]
R11	$CH_3 + O_2 = CH_2O + OH$	6.38×10^{11}	0.00	13,594	[49]
R12	$CH_3 + O_2 = CH_3O + O$	7.55×10^{12}	0.00	28,320	[48]
R13	$CH_3 + O_2$ (+M) = CH_3O_2 (+M)	7.81×10^{9}	0.90	0.00	[50]
	Low /	6.85×10^{24}	−3.00	0.00/	
	Troe /	0.60 1000 70 1700/			
R14	$CH_3 + O = CH_2O + H$	3.05×10^{13}	0.05	−136	[52,57]
R15	$CH_3 + O = CO + H + H_2$	2.49×10^{13}	0.05	−136	[52,57]
R16	$CH_3 + HO_2 = CH_3O + OH$	1.00×10^{12}	0.269	−687.5	[59]
R17	$CH_3 + HO_2 = CH_4 + O_2$	1.16×10^{5}	2.23	−3022	[59]
	REV/	1.30×10^{6}	2.412	51,288/	[62]
R18	$CH_3 + CH_3$ (+M) = C_2H_6 (+M)	8.88×10^{16}	−1.16	774.5	[64]
	Low /	3.74×10^{50}	−9.93	7389/	
	Troe /	0.75 158 32,828 46,564/			
	H_2/3.0/ H_2O/9.0/ CO/2.25/ CO_2/3.0/ Ar/1.0/ CH_4/3.0/ C_2H_6/4.5/ N_2/1.5/				
R19	HCO (+M) = H + CO (+M)	4.93×10^{16}	−0.93	19,724	[65]
	Low /	7.43×10^{21}	−2.36	19,383/	
	Troe /	0.103 139 10,900 4550/			
	H_2/2.0/ H_2O/15.0/ CO/1.5/ CO_2/3.0/ Ar/1.0/ He/1.5/ O_2/1.5/ CH_4/5.0/ N_2/1.5/				
R20	CO + O (+M) = CO_2 (+M)	1.80×10^{10}	0.0	2384	[20]
	Low /	1.55×10^{24}	−2.79	4191/	
	H_2/2.5/ H_2O/12.0/ CO/1.9/ CO_2/3.8/ Ar/0.87/				
R21	$H + O_2 = O + OH$	1.04×10^{14}	0.0	15,286	[66]
R22	$H + O_2$ (+M) = HO_2 (+M)	4.65×10^{12}	0.44	0.00	[67]
	Low /	6.37×10^{20}	−1.72	524.8/	
	Troe /	0.5 1×10^{-30} 1×10^{30}/			
	H_2/2.0/ H_2O/14.0/ CO/1.9/ CO_2/3.8/ Ar/0.67/ He/0.8/ O_2/0.78/				

taken from literature work are presented in Table S1 in the *Supplementary materials* while the comparisons between these literature data and the simulated results of the present model are shown in Figs. S1–S9 in the *Supplementary materials*.

4. Results and discussion

In this section, the performance of the present model against previous and present laminar flame speed targets is examined at first. Then the contribution of thermal and chemical effects to the variation of the laminar flame speed by adding CO is discussed at different equivalence ratios and pressures. In the end, the transition from CH_4-related chemistry to CO-related chemistry under different conditions is compared and clarified based on the model analysis.

4.1. Model validation against laminar flame speed targets

4.1.1. Hydrogen/syngas flames

Since the incorporation of R1–R4 could alter the performance of the base model, previous validation targets are used to re-examine the base model. According to our validation results, the validation targets of CH_2O and CH_3OH are not sensitive to R1–R4 while those of H_2 or H_2-enriched fuels are sensitive to them. In addition, the experimental data obtained under ignition conditions are not sensitive to R1–R4. Only under flame conditions, where the temperature is high and radicals are abundant, R1–R4 can play an important role. Figure 1 presents the comparison between the measured and predicted laminar flame speeds of H_2/air mixtures at 298 K and 1 atm. As we can see, incorporating R1–R4 in the present model could decrease the predictions evidently. Besides, in syngas flames, similar influence of R1–R4 on the predictions of laminar flame speeds can be observed, as shown in Fig. 2. In general, the present model is still adequate to predict the flame speed targets in literature. The decrease of the laminar flame speeds with the incorporation of R1-R4 in the present model is due to the loss of H atom, which has a high thermal diffusion coefficient and chemical reactivity and plays an important role in determining the laminar flame speed.

The present model was also validated against the laminar flame speed data of CH_4/air mixtures under various pressure conditions. The predicted results of the present model against the experimental targets of CH_4/Air mixture at 1–10 atm are presented in Figs. S1–S4 in the *Supplementary materials*. In general, the present model could reasonably predict all the experimental data.

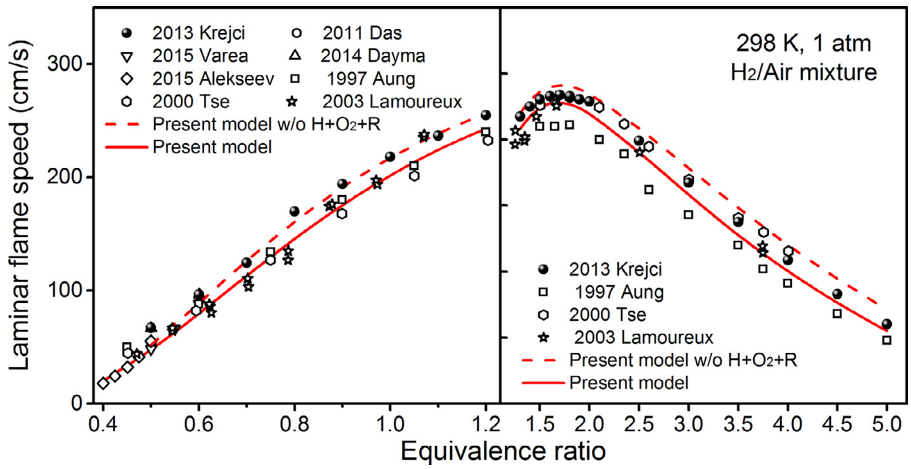

Fig. 1. Measured (symbols) [15,72–79] and predicted (lines) laminar flame speeds of H_2/air mixtures at 298 K and 1 atm. Solid and dashed lines are the predicted results of the present model with and without R1–R4, respectively.

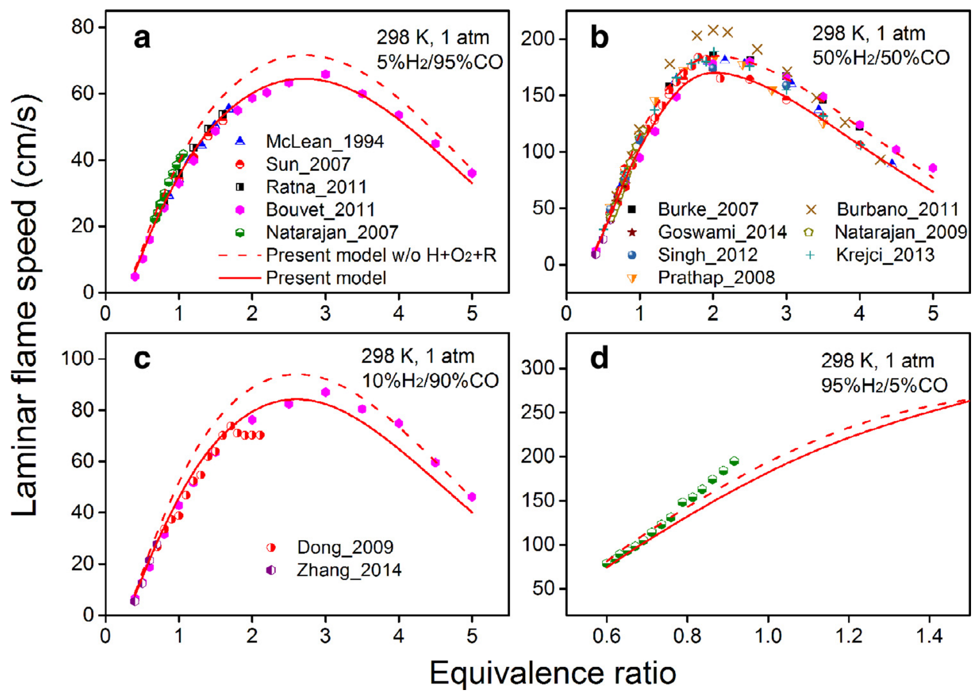

Fig. 2. Measured (symbols) [76,80–92] and predicted (lines) laminar flame speeds of syngas/air mixture at 298 K and 1 atm. Solid and dashed lines are the predicted results of the present model with and without R1–R4, respectively.

4.1.2. Methane/carbon monoxide flames

The measured laminar flame speeds of CH_4/CO/air mixtures at 353 K and 1–2 atm are shown in Fig. 3. Similar to previous experimental studies, the laminar flame speed first increases and then decreases with the increasing CO content in the mixture. The peak value locates at around 90% of CO content in the CH_4/CO mixture. The comparison between results at different equivalence ratios and same pressure shows that the increment of laminar flame speed under the rich conditions is more visible. Figures 4(a,b) and 5(a,b) present the measured laminar flame speeds of CH_4/CO/air mixtures at 5 and 10 atm, respectively, while Figs. 4(c,d) and 5(c,d) are the results of CH_4/CO/O_2/He mixtures at 5 and 10 atm, respectively. Since the flame instability of pure CH_4 flame becomes severe under high pressure conditions which may introduce large experimental uncertainties, the experimental data of pure CH_4 flames at 5 and 10 atm are not presented in this work. Similar to the situations at 1 and 2 atm, the increments of the laminar flame speeds under the rich conditions are also more pronounced than those under the lean conditions at 5 and 10 atm. Besides, the peak value of the laminar flame speed also appears at around 90% of CO content in the CH_4/CO mixture at 5 and 10 atm.

The predicted results of the present model, as well as previous models [69–71], are also shown in Figs. 3–5. As we can see, the present model could reasonably predict the distribution of the laminar flame speed with the increasing CO content under different pressure and equivalence ratio conditions. Among the four models, the present model has better performance under lean conditions while the FFCM-1 model [70] can better capture the experimental data under rich conditions. The HP model [69] and the San Diego model [71] can better capture the peak laminar flame speeds while the former one suffers from the convergence problems at 10 atm. Besides, the present model was also validated against the laminar flame speed data of CH_4/CO/air mixtures at ambient conditions in literature [7–9], as shown in Figs. S5–S9 in the *Supplementary materials*. In general, the present model could reasonably predict both the present and literature laminar flame speed data

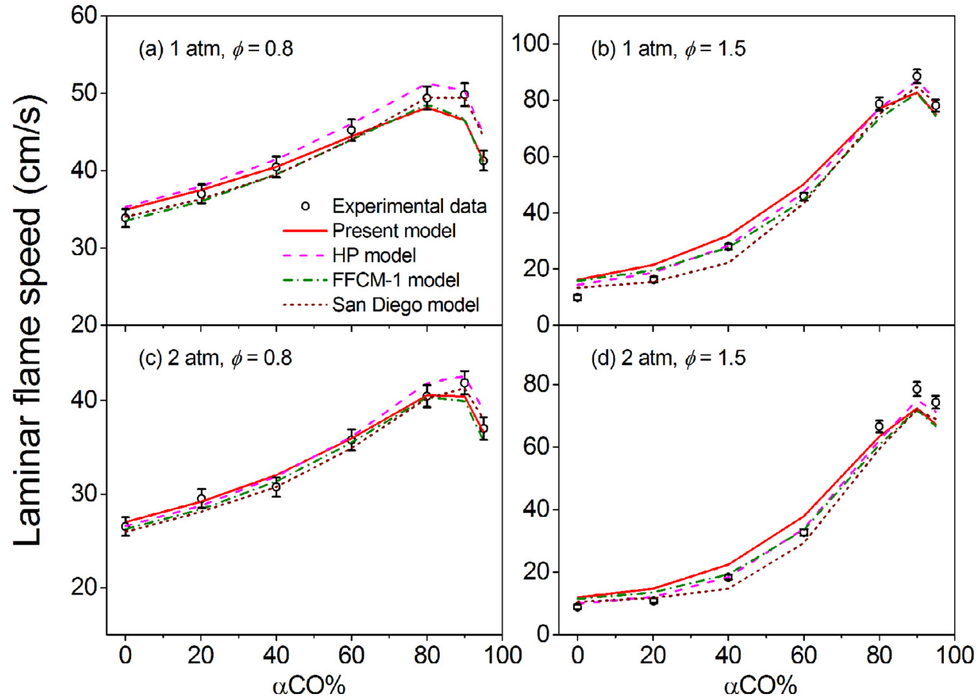

Fig. 3. Measured (symbols) and predicted (lines) laminar flame speeds of CH_4/CO/air mixtures at 353 K, 1–2 atm and equivalence ratios of 0.8 and 1.5. Solid, dashed, dash dotted and short dashed lines represent the predicted results of the present model, HP model [69], FFCM-1 model [70] and San Diego model [71], respectively.

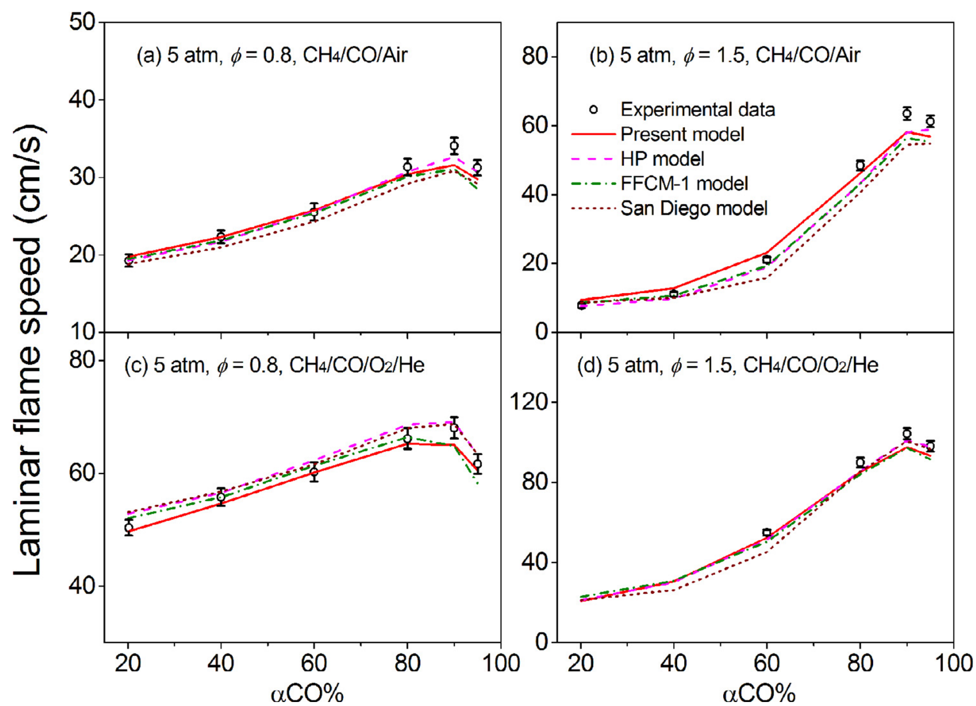

Fig. 4. Measured (symbols) and predicted (lines) laminar flame speeds of (a,b) CH_4/CO/air and (c,d) $CH_4/CO/O_2/He$ mixtures at 353 K, 5 atm and equivalence ratios of 0.8 and 1.5. Solid, dashed, dash dotted and short dashed lines represent the predicted results of the present model, HP model [69], FFCM-1 model [70] and San Diego model [71], respectively.

of CH_4/CO/air mixtures with the consideration of experimental uncertainties.

4.2. Analysis on the effects of CO addition

CO addition could increase the laminar flame speed of methane flames according to previous study by Wu et al. [12]. Based on the present model, we calculated the percentage increase in laminar flame speed ($\Delta S_u\%$) caused by CO addition compared with the CH_4/air mixture under different conditions according to Eq. (1), where $S_u(CH_4+CO)$ and $S_u(CH_4)$ represent the predicted laminar flame speed of CH_4/CO/air mixture and CH_4/air mixture based on the present model, respectively. The calculated results are plotted in Fig. 6 (blue filled bars).

$$\Delta S_u\% = \frac{[S_u(CH_4+CO) - S_u(CH_4)]}{S_u(CH_4)} \times 100\% \tag{1}$$

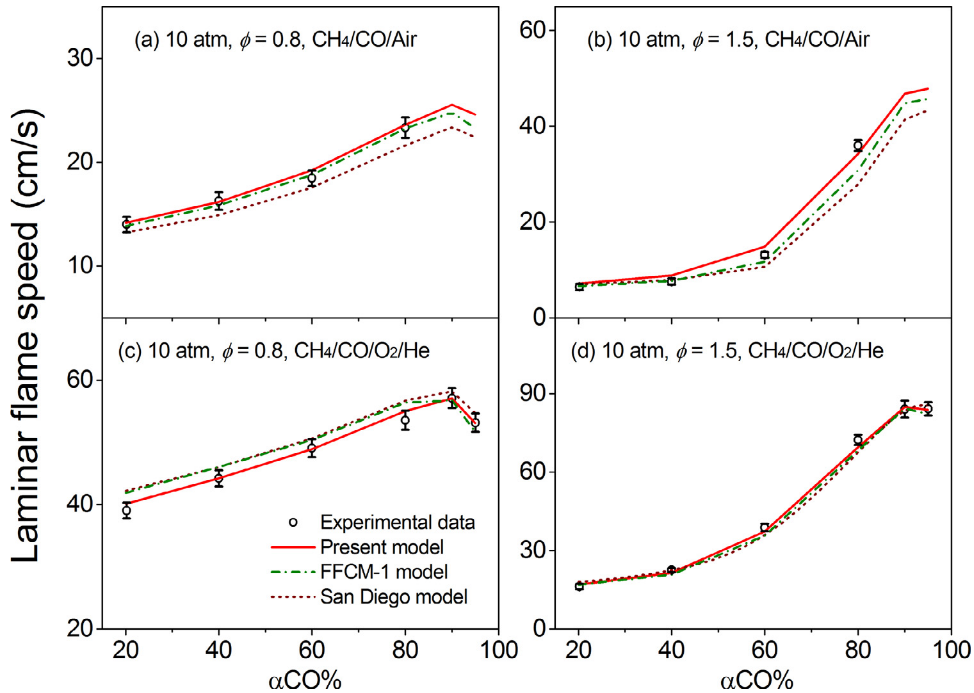

Fig. 5. Measured (symbols) and predicted (lines) laminar flame speeds of (a,b) CH_4/CO/air and (c,d) CH_4/CO/O_2/He mixtures at 353 K, 10 atm and equivalence ratios of 0.8 and 1.5. Solid, dash dotted and short dashed lines represent the predicted results of the present model, FFCM-1 model [70] and San Diego model [71], respectively.

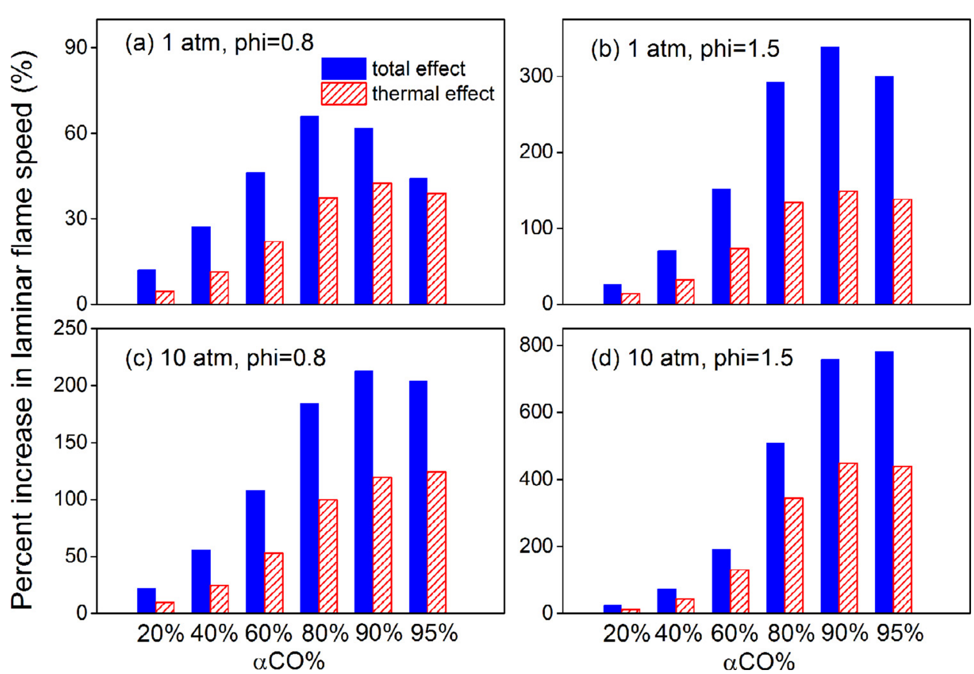

Fig. 6. Total effects and thermal effects on laminar flame speed with CO addition, at $\phi = 0.8$ and 1.5 and pressures of 1 and 10 atm. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

It is found that the rich conditions at high pressure (see Fig. 6(d)) are the most favorable conditions, especially with high CO contents. Normally, the increase of the laminar flame speed is mainly considered to be resulted from thermal, transport and chemical effects. Among them, the thermal effect is mainly related to the variation in adiabatic flame temperature and the chemical effect is closely related to the variation in key radicals. The transport effect plays a minor role compared with the other two effects according to previous studies [10,93], which is related to the variation in thermal and mass diffusivity (i.e. Lewis number).

In this work, we first isolate the thermal effect from the total effect (referred as Step 1) with the same method mentioned in [11,93], which is introduced in detail in [94]. In brief, the compositions of CH_4/CO/air mixtures are adjusted by replacing part of nitrogen with fake nitrogen (FN_2) in order to reduce the flame temperatures to reach the adiabatic flame temperature of the CH_4/air mixture. Here the thermal and transport properties of the FN_2 are equal to carbon dioxide and nitrogen, respectively. The FN_2 is treated as an inert gas and is assigned as the same third-body collision enhancement factor as nitrogen. The mole fraction of the inert gas is kept constant. In this way, the thermal effect can be observed by comparing the difference of the laminar flame speed between the real (CH_4/CO/air) and fake ($CH_4/CO/O_2/N_2/FN_2$) mixtures, as shown in Fig. S10 (squares and circles). The thermal effect on the $\Delta S_u\%$ is shown in Fig. 6 (red oblique line bars). The calculation method is also based on Eq. (1), where $S_u(CH_4+CO)$ and $S_u(CH_4)$ represent the predicted laminar flame speed of $CH_4/CO/O_2/N_2/FN_2$ mixture and CH_4/air mixture based on the present model, respectively. As we can see, at 1 atm, compared with the rich conditions, the thermal effect plays a more important role under the lean conditions, especially when the CO content in the mixture is higher than 60%.

In order to evaluate the transport effect, based on the $CH_4/CO/O_2/N_2/FN_2$ mixture, the transport properties of CO were artificially assigned equal to CH_4, O_2 and N_2, separately (referred as Step 2). It can be found that the predicted laminar flame speeds in this way change little, as seen from Fig. S10 (circles and stars). Therefore, the transport effect only has very weak influence under the investigated conditions, which is in accordance with previous conclusions [10,93]. The difference between the total and thermal effect shown in Fig. 6 could roughly represent the contribution of chemical effect. As we can see, the chemical effect plays an important role under the rich conditions at 1 atm, especially with high CO contents. The present results of the effect of CO addition are compared with those obtained in previous studies [10–12]. Unlike the conclusion drawn by Wu et al. [12] that the chemical effect dominates the transition chemistry, in the present work, it is found that the thermal effect can also play an important role, especially under the lean conditions at 1 atm, as can be seen from Fig. 6(a). Compared with the conclusions drawn by Sung et al. [10] in *n*-butane/CO/air mixtures and Liu et al. [11] in CH_4/syngas/air mixtures that the thermal effect mainly accounts for the increment of the laminar flame speed, it is found that their conclusion is only suitable for some limited conditions in the present CH_4/CO/air mixtures. As shown in Fig. 6, the chemical effect can be comparable with the thermal effect under most investigated conditions. Therefore, based on the present work, it is found that both the thermal effect and chemical effect can play important roles in the increase of the laminar flame speed simultaneously. The separate contribution of each effect varies at different pressures, equivalence ratios and CO contents.

4.3. Transition chemistry from CH_4 to CO

4.3.1. Change of the radical pool

In order to reveal the transition from CH_4-related chemistry to CO-related chemistry under different investigated conditions, the radical pool under both the lean and rich conditions at 1 and 10 atm is plotted in Fig. 7. Since we focus on the transition chemistry, CO contents from 60% to 95% in CH_4/CO/air mixtures were selected for analyses based on the comparison with CH_4/air mixtures. In CH_4/air mixtures, OH radical is the most abundant radical in the radical pool under the lean conditions while H atom and CH_3 radical are more important under the rich conditions. However, the concentrations of these radicals changes gradually with the increasing CO content. Particularly, the mole fraction of CH_3 radical becomes less while that of O atom increases with the increasing CO content. Since CH_3 radical is the dominant decomposition product of methane, it can be considered as an indicator of methane chemistry. Meanwhile, O atom is the only radical in dry CO mechanism, which can be an indicator of CO mechanism. The decreasing trend of CH_3 and increasing trend of O atom observed in the radical pool under various conditions suggest that the dominant chemistry is shifting from CH_4 to dry CO. Furthermore, when the CO content reaches 95% under the lean conditions, OH radical is no longer the dominant radical in the radical pool, while O atom replaces its role. Similarly, under the rich conditions, the important role of CH_3 in CH_4/air mixture is also replaced by O atom when the CO content is high enough.

4.3.2. Changes of the reaction pathways

In order to reveal the transition from CH_4-related chemistry to CO-related chemistry, main reaction pathways are analyzed based on the rate of production (ROP) analysis. As a result, the reaction network is illustrated in Fig. 8. In both the CH_4/air and CH_4/CO/air flames, methane is mainly consumed via the H-abstraction reactions by H, O and OH radicals (R7–R9) to form CH_3 radical. CH_3 radical has various consumption pathways in the CH_4/air flames, mainly including R14, R15, R6 and R18. Among them, R14 and R15 are the bimolecular reactions between CH_3 and O atom, which are more important under the lean conditions. In contrast, R6 and R18 have larger contributions under the rich conditions since H atom and CH_3 radical are abundant under these conditions. With the increasing CO content, the contribution of R18 becomes weaker, and thus the subsequent C_2 pathways become less important. When the CO content is equal to 95%, the contribution of R18 can be neglected. The decreasing contribution of C_2 pathways with the increasing CO content is an important feature in the transition from CH_4-related chemistry to CO-related chemistry, which is highlighted in Fig. 8 with blue box.

$$CH_3 + CH_3\ (+M) = C_2H_6\ (+M) \tag{R18}$$

Formaldehyde (CH_2O) is generated via R14, which subsequently reacts with various radicals to form HCO radical. The main consumption pathways of HCO radical are listed in Fig. 8, all of which contribute to the formation of CO. It is noted that in the CH_4/air flames, R19 proceeds in its forward direction, releasing H atom which promotes the flame propagation. With the increasing CO content in the CH_4/CO/air mixtures, the contribution of R19 becomes less. When the CO content is equal to 95%, R19 proceeds in its reverse direction, consuming H atom and consequently inhibiting the flame propagation. For the consumption of CO, in the CH_4/air flames, CO is dominantly consumed via R5 while the contribution of R5 becomes less in the CH_4/CO/air flames. At the same time, the contribution of R20 increases with the increasing amount of O atom (see Fig. 7).

$$HCO\ (+M) = CO + H\ (+M) \tag{R19}$$

$$CO + O\ (+M) = CO_2 + (+M) \tag{R20}$$

Since H atom is the most important radical in flame propagation through the crucial chain-branching reaction (R21), its consumption pathways are compared between the CH_4/air and CH_4/CO/air flames. In the CH_4/air flames, H atom can be consumed via R6 and R7. R6 is a chain-termination reaction that inhibits the flame propagation. In contrast, with the increasing CO content, the consumption of H atom via R6 and R7 becomes less while R21 becomes more important for consuming H atom. In this way, the inhibition effect of R6 on the flame propagation becomes weak while the promotion effect of R21 is enhanced.

$$H + O_2 = O + OH \tag{R21}$$

4.3.3. Effects of equivalence ratios and pressures

In this work, the transition from CH_4-related chemistry to CO-related chemistry is found to be influenced by equivalence ratios and pressures. As mentioned above, the increasing CO content leads to enhanced O atom production (see Fig. 7). In CH_4/air flames, the contributions of R7–R9 to the consumption of CH_4 have large differences under the lean and rich conditions. Under the

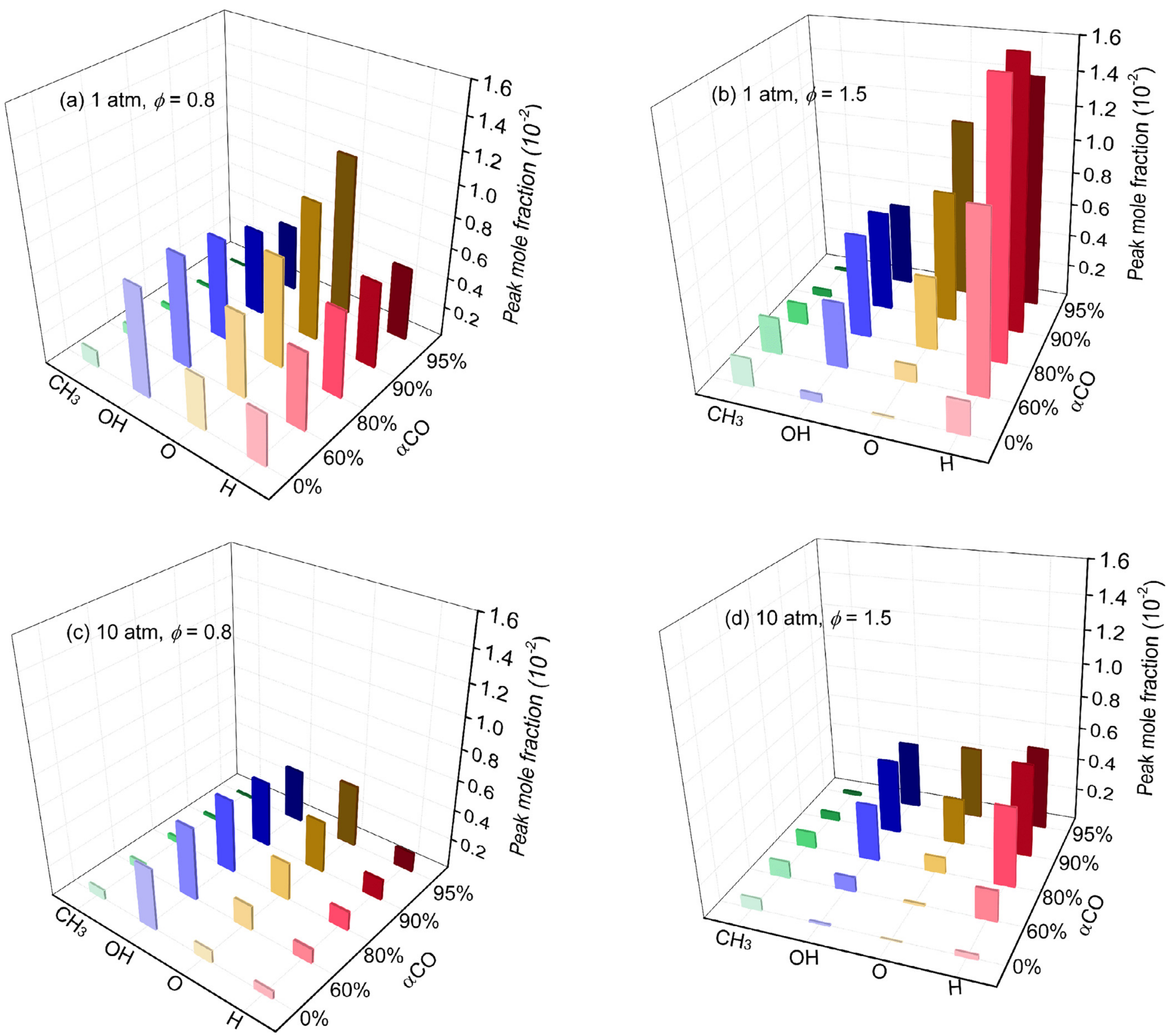

Fig. 7. Radical pool (H, O, OH and CH_3) in the CH_4/air flames and CH_4/CO/air flames with the CO content of 60%, 80%, 90% and 95%.

lean conditions, the contribution of R9 is bigger than R7, while R7 is more important under the rich conditions. The contribution of R8 plays a minor role to the consumption of CH_4 under both the lean and rich conditions. However, with the increasing CO content, R8 becomes more and more important. This is due to the increasing contribution of R5. R5 converts OH radical to H atom, and H atom is then consumed via R21 to produce O atom and OH radical. In this reaction circle, OH radical and H atom are kept invariable while O atom is accumulated. Therefore, with the increasing CO content, the concentration of O atom rapidly increases, which leads to the increasing contribution of R8.

As discussed above, it is found that the chemical effect plays a more important role in the increase of laminar flame speed under the rich conditions at 1 atm, as shown in Fig. 6(a,b). Based on the above modeling analysis, R18, R19, R6 and R7 have major contributions to the transition from CH_4-related chemistry to CO-related chemistry. R6 and R18 are chain-termination reactions and R7 converts reactive H atom to less reactive CH_3 radical, all of which inhibit the flame propagation. Since these reactions are more important under the rich conditions, their decreasing contributions in the transition from CH_4-related chemistry to CO-related chemistry could increase the laminar flame speeds more dramatically under the rich conditions. As for R19, under the rich conditions, the concentrations of CO and H atom are more considerable than those under the lean conditions. Thus the inhibiting effect of R19 on the flame propagation is more severe under the rich conditions than the lean conditions.

For the pressure effects, it is also observed from Fig. 6 that the relative increase in laminar flame speed at 10 atm due to CO addition is more remarkable than that at 1 atm, which is mainly caused by the influence of pressure-dependent reactions. For example, R22 is the competitive reaction of the crucial chain-branching reaction (R21) and high pressure conditions can make R22 more competitive. Therefore, under high pressure conditions, the production of O atom via R21 would be reduced. Besides, the contribution of R20 will also increase with the increasing pressure, which leads to more consumption of O atom. Based on the above reasons, the net production rate of O atom at high pressures decreases, which can be seen from Fig. 7. Besides, R6 and R18 are chain-termination reactions and inhibit the flame propagation.

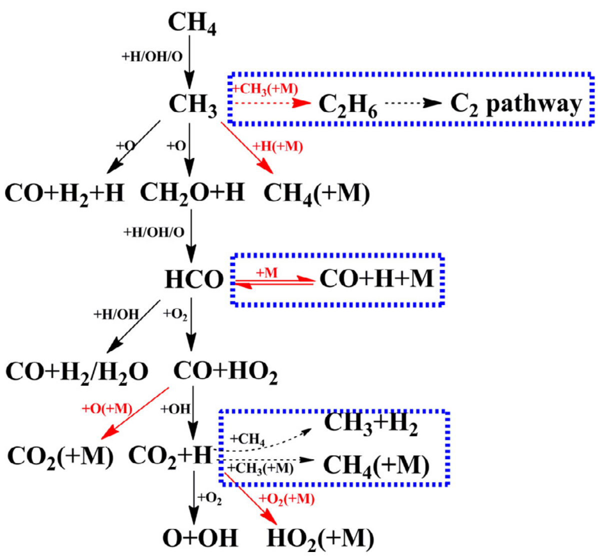

Fig. 8. Reaction network based on the ROP analysis in the CH_4/CO/air flames. Reaction pathways contributing to the transition from CH_4-related chemistry to CO-related chemistry are highlighted in blue boxes. Pressure-dependent pathways are highlighted with red arrows. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The decreasing contribution of R6 and R18 in the transition from CH_4-related chemistry to CO-related chemistry can increase the laminar flame speed. R6 and R18 are pressure-dependent reactions, and with the increase of pressure, the contribution of R6 and R18 becomes more important. Therefore, the relative increase of the laminar flame speed under high pressure condition due to CO addition becomes more evident than that at 1 atm.

$$H + O_2\ (+M) = HO_2\ (+M) \tag{R22}$$

5. Conclusions

Laminar flame speeds of CH_4/CO/air and CH_4/CO/O_2/He mixtures were measured at an unburned temperature of 353 K and a pressure range of 1–10 atm in a high-pressure constant-volume cylindrical combustion vessel. The equivalence ratios of 0.8 and 1.5 were selected to represent the lean and rich conditions, respectively. The molar percentage of CO varies from 0% to 95% at 1 and 2 atm and 20% to 95% at 5 and 10 atm. Besides, a kinetic model for CH_4/CO mixture was developed and validated against the laminar flame speed data in the present work. Based on the modeling analysis, it is found that both the thermal effect and chemical effect play important roles in the variation of laminar flame speed. The separate contribution of each effect varies at different pressures, equivalence ratios and CO contents. Besides, the transition from CH_4-related chemistry to CO-related chemistry can be monitored by the increasing amount of O atom in the radical pool under all the investigated conditions. Based on the modeling analysis, R18, R19, R6 and R7 have major contributions to the transition chemistry. Among them, the contributions of R18, R6 and R7 become less important with the increasing CO content while that of R19 proceeds in reverse direction. These reactions are more important under the rich conditions. Therefore, their contributions to the transition chemistry under the rich conditions are larger than those under the lean conditions. Compared with atmospheric pressure conditions, the contributions of R20 and R22 are enhanced at high pressures, which leads to the decrease of O atom in the radical pool. Besides, since the chain-termination reactions R6 and R18 become more important as the pressure increases, their decreasing contribution in the transition process from methane-related chemistry to CO-related chemistry leads to the more evident increase of laminar flame speed with CO addition at high pressures.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.08.021.

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